

Linear models are one of the simplest and most well-understood classes of models.

Definition:

$$Y = X\beta + \varepsilon$$

Annotations:

- Y : observations
- X : design matrix
- β : parameter vector
- ε : error vector such that $E(\varepsilon) = 0$

If $\varepsilon \sim N(0, \sigma^2 I)$, then the linear model is normal linear model and we have

$$Y|_X \sim N(X\beta, \sigma^2 I)$$

Annotations:

- 1. uncorrelated
- 2. Same Variance

Estimation of β :

We can consider estimating β for different perspectives and they all lead to the same result.

1. minimize MSE

$$\begin{aligned} e &= Y - \hat{Y} = Y - X\beta \\ \Rightarrow e^T e &= (Y - X\beta)^T (Y - X\beta) \\ \Rightarrow \frac{\partial(e^T e)}{\partial \beta} &= 2(Y - X\beta)^T (-X) \\ &= -2(Y^T X - \beta^T X^T X) \end{aligned}$$

$$\text{Let } Y^T X = \beta^T X^T X$$

$$\Rightarrow \beta^T = Y^T X (X^T X)^{-1} \quad (\text{if } X \text{ has full column rank})$$

$$\Rightarrow \beta = (X^T X)^{-1} X^T Y$$

2. maximize likelihood function

$$\mathcal{L}(\beta, \sigma^2; Y) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2} (Y - X\beta)^T (Y - X\beta)\right\}$$

$$\Rightarrow \mathcal{L}(\beta, \sigma^2; Y) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (Y - X\beta)^T (Y - X\beta)$$

$\Rightarrow$ maximize likelihood function $\Leftrightarrow$ minimize MSE

3. projection

Consider solve $Y = X\beta$, we can not find a solution for β

because normally $n \gg m$ so $Y \notin \text{Col}(X)$

Therefore, we choose a $\hat{\beta}$ such that $X\hat{\beta}$ is the closest vector to Y

this vector is project of Y in X

$$\Rightarrow X(X^T X)^{-1} X^T Y = X\hat{\beta}$$

$$\Rightarrow \hat{\beta} = (X^T X)^{-1} X^T Y$$

If $(X^T X)^{-1}$ does not exist, in other words X does not have full column rank
e.g. if $m > n$

1. Ridge Regression

$$\begin{aligned}\hat{\beta} &= (X^T X + \lambda I)^{-1} X^T y \\ &\downarrow \\ &= U(\Lambda + \lambda I)U^T \\ &\downarrow \\ &\text{all eigenvalues } > \lambda\end{aligned}$$

2. pseudoinverse of X

Any A has SVD: $A = U \Sigma V^T$

A^+ is defined as: $A^+ = V \Sigma^+ U^T$

$\downarrow$
take transpose of Σ
and let $\sigma_i \rightarrow \frac{1}{\sigma_i}$

1. if A^{-1} exists, then $A^+ = A^{-1}$

2. if $(A^T A)^{-1}$ exists then $A^+ = (A^T A)^{-1} A^T$

$$\begin{aligned}&\rightarrow \frac{(V \Sigma^T U^T U \Sigma V^T)^{-1} V \Sigma^T U^T}{\text{Compact version of SVD}} \\ &= (V \Sigma^T V^T)^{-1} V \Sigma^T U^T \\ &= V \Sigma^{-2} V^T V \Sigma U^T \\ &= V \Sigma^{-1} U^T\end{aligned}$$

3. if $(A^T A)^{-1}$ does not exist

then $A^T A \beta = A^T y$ has infinite solutions

because $\text{Col}(A^T A) = \text{Col}(A^T)$

$\hat{\beta} = A^+ y$ leads to solution with minimum L_2 norm (i.e. $\|\hat{\beta}\|_2$)

Consider another solution β^* , we have

$$A^T A (\hat{\beta} - \beta^*) = 0$$

$$\Rightarrow A^T A \hat{\beta} - A^T A \beta^* = 0$$

$$\Rightarrow \beta^* = \hat{\beta} + n \rightarrow n \in \text{Null}(A^T A) = \text{Null}(A)$$

$$A^T y = \underbrace{V \Sigma^T U^T y}_{\in \text{Col}(A^T)} \text{ so } \in \text{Col}(A^T A)$$

$$\begin{aligned} \text{Since } \hat{\beta} \perp n \text{ then } \beta^* \cdot \beta^* &= (\hat{\beta} + n) \cdot (\hat{\beta} + n) \\ &= \|\hat{\beta}\|_2^2 + \|n\|_2^2 \geq \|\hat{\beta}\|_2^2 \end{aligned}$$

For the reasons above, for solving $Ax = \beta$, we often just use $x = A^+ \beta$ directly.

properties of $\hat{\beta}$:

We know $\hat{\beta} = (X^T X)^{-1} X^T Y$ (given that X has full column rank)

Since $Y|X \sim N(X\beta, \sigma^2 I)$ and $\hat{\beta}$ is a linear combination of Y , $\hat{\beta}$ must follow normal distribution as well.

Further, we have

$$E(\hat{\beta}) = E((X^T X)^{-1} X^T Y)$$

$$= (X^T X)^{-1} X^T E(Y)$$

$$= (X^T X)^{-1} X^T X \beta$$

$$= \beta$$

$\Rightarrow$ So $\hat{\beta}$ is an unbiased estimation of β

$$\begin{aligned} \text{Cov}(\hat{\beta}) &= \text{Cov}((X^T X)^{-1} X^T Y) = (X^T X)^{-1} X^T \text{Cov}(Y) X (X^T X)^{-1} \\ &= \sigma^2 (X^T X)^{-1} \end{aligned}$$

Gauss - Markov Theorem: assume a linear model with unknown parameter β , a design matrix with full rank and $\text{Cov}(\epsilon) = \sigma^2 I$. For any fixed $C \in \mathbb{R}^m$, let $\hat{\beta}$ be the least square estimator of the linear model, then $C^T \hat{\beta}$ has the smallest variance among all linear unbiased estimator for $C^T \beta$.

Suppose there is another linear, unbiased estimator $\tilde{\beta}$. Since it is linear, we must have

$$\tilde{\beta} = WY$$

$$\text{Let's define } W = (X^T X)^{-1} X^T + D$$

Since $\tilde{\beta}$ is unbiased, we have

$$\begin{aligned} E(\tilde{\beta}) &= E(WY) = E(W(X\beta + \epsilon)) \\ &= WX\beta \end{aligned}$$

$$\Rightarrow WX = I$$

$$\Rightarrow DX = 0$$

$$\Rightarrow \text{Var}(\tilde{\beta}) = \text{Var}(WY) = W \text{Var}(Y) W^T = \sigma^2 W W^T$$

$$= \sigma^2 ((X^T X)^{-1} X^T + D) ((X^T X)^{-1} X^T + D)^T$$

$$= \sigma^2 ((X^T X)^{-1} + DD^T)$$

$$= \text{Var}(\hat{\beta}) + \sigma^2 DD^T$$

$$\Rightarrow \text{Var}(C^T \tilde{\beta}) = C^T \text{Var}(\tilde{\beta}) C + \sigma^2 C^T D D^T C$$

$$\geq \text{Var}(C^T \hat{\beta})$$

Estimation of σ^2

Let $e = (Y - \hat{Y})$, $RSS = e^T e$, then $\hat{\sigma}^2 = \frac{RSS}{n-r}$ is an unbiased estimation of σ^2

$$\begin{aligned} E(RSS) &= E((Y - \hat{Y})^T (Y - \hat{Y})) \\ &= E((Y - PY)^T (Y - PY)) \quad \text{where } P = X(X^T X)^{-1} X^T \\ &= E(Y^T (I - P)^T (I - P) Y) \end{aligned}$$

P is a projection matrix $\Rightarrow I - P$ is also a projection

$$\Rightarrow (I - P)^T = (I - P) \text{ and } (I - P)^2 = (I - P)$$

$$\text{So } E(RSS) = E(Y^T (I - P) Y)$$

$$\begin{aligned} \text{Since } (I - P)^T &= I - P, \text{ we must have } I - P = S \Lambda S^T \\ &\Rightarrow (I - P)^2 = S \Lambda^2 S^T \\ &\Rightarrow \text{eigenvalue of } I - P = 0 \text{ or } 1 \end{aligned}$$

$$\begin{aligned} \text{Since Sum of eigenvalues} &= \text{trace}(I - P) \\ &= n - \text{trace}(A(A^T A)^{-1} A^T) \\ &= n - \text{trace}(A^T A (A^T A)^{-1}) \\ &= n - m \end{aligned}$$

$\Rightarrow \Lambda$ has $n - m$ 1 on its diagonal

$$\Rightarrow E(RSS) = E(Y^T S \Lambda S^T Y)$$

$$\begin{aligned} \text{Let } Z &= S^T Y, \text{ we have } \text{Cov}(Z) = S^T \text{Cov}(Y) S \\ &= \sigma^2 I \end{aligned}$$

$\Rightarrow Z$ is independent normal distributions with variance σ^2

$$\Rightarrow \frac{Z^T \Lambda Z}{\sigma^2} \sim \chi^2_{n-m}$$

$$\Rightarrow E\left(\frac{Z^T \Lambda Z}{\sigma^2}\right) = n - m$$

$$\Rightarrow E\left(\frac{Z^T \Lambda Z}{n - m}\right) = \sigma^2$$

Inference

We now consider the Confidence interval for an estimation $C^T \beta$

$$\text{We know } \hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$$

$$\Rightarrow C^T \hat{\beta} \sim N(C^T \beta, \sigma^2 C^T (X^T X)^{-1} C)$$

$$\Rightarrow \frac{C^T \hat{\beta} - C^T \beta}{\sqrt{\sigma^2 C^T (X^T X)^{-1} C}} \sim N(0, 1)$$

plug in $\hat{\sigma}^2 = \frac{RSS}{n-p}$

$$\Rightarrow \frac{C^T \hat{\beta} - C^T \beta}{\sqrt{\hat{\sigma}^2 C^T (X^T X)^{-1} C}} \sim t_{n-p}$$

$$\Rightarrow P(t_{n-p}(1-\alpha/2) \leq \frac{C^T \hat{\beta} - C^T \beta}{\sqrt{\hat{\sigma}^2 C^T (X^T X)^{-1} C}} \leq t_{n-p}(\alpha/2)) = \alpha$$

$$\Rightarrow P(C^T \beta \in [C^T \hat{\beta} - t_{n-p}(\alpha/2) \sqrt{\hat{\sigma}^2 C^T (X^T X)^{-1} C}, C^T \hat{\beta} + t_{n-p}(\alpha/2) \sqrt{\hat{\sigma}^2 C^T (X^T X)^{-1} C}]) = \alpha$$

Note: 1. if we want to have Confidence interval of β_i ,

just let $C = (0, 0, \dots, 1, 0, \dots, 0)$
act component i

2. C^T can be a input, in this case

the Confidence interval above is for $C^T \beta$ (average output given input)

if we want Confidence interval for a single estimation,

need to take it into consideration